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7.1 History of a Cell

Table 7.1: History of Cell		
Year	**Scientist**	**Events**
1665	Robert Hooke	Coined word "Cell"
1670–74	Antonie van Leeuwenhoek	First living cells observed in microscope - Structure of bacteria
1831–33	Robert Brown	Presence of nucleus in cells of orchid roots
1839	Jan Evangelista Purkyne (J.E. Purkinje)	Coined "protoplasm"
1838–39	Schleiden & Schwann	Cell theory
1858	Rudolph Ludwig Carl Virchow	Cell theory '*omnis cellula e cellula*'
1873	Anton Schneider	Described chromosomes (Nuclear filaments) for the first time
1882	Walther Flemming	Coined the word mitosis; chromosome behaviour
1883	Edouard Van Beneden	Cell division in round worm
1888	Theodor Boveri	Centrosome; Chromosome Theory

7.1.1 The Role of the Nucleus

As studied earlier, the nucleus is the organising centre of the cell. The information in the nucleus is contained within structures called **chromosomes**. These uniquely:

- Control activities of the cell.
- Genetic information copied from cell to cell while the cell divides.
- Hereditary characters are passed on to new individuals when gametic cells fuse together in sexual reproduction.

7.1.2 Chromosomes

At the time when a nucleus divides, the chromosomes become compact and multicoiled structure. Only in this condensed state do the chromosomes become clearly visible in cells. All other times, the chromosomes are very long, thin, uncoiled threads. In this condition they give the stained nucleus the granular appearance. The granules are called **chromatin**.

The four important features of the chromosome are:

- **The shape of the chromosome is specific:** The long, thin, lengthy structured chromosome contains a short, constricted region called **centromere**. A centromere may occur anywhere along the chromosome, but it is always in the same position on any given chromosome.
- **The number of chromosomes per species is fixed:** for example the mouse has 40 chromosomes, the onion has 16 and humans have 46.